

Simultaneous Multislice Reconstruction

Steen Moeller^a and Suchandrima Banerjee^b

^aCenter for Magnetic Resonance Research, University of Minnesota, Minneapolis, MN, United States

^bGE Healthcare, Menlo Park, CA, United States

7.1 Introduction

Shorter MR scan times are desirable from considerations of scan throughput, patient comfort, and reducing incidences of motion. Parallel imaging accelerates an MR scan by acquiring fewer number of phase-encode steps, at less dense spacing than needed to meet the Nyquist criterion and avoid aliasing, while keeping the spatial frequency extent unchanged. The missing information is derived by exploiting the additional spatial encoding provided by signal reception with multiple coil elements [23,56,63]. Other sparse data sampling approaches have also been introduced to shorten the scan time [35]. These techniques have been successfully applied to speed up two dimensional (2D) and three dimensional (3D) Cartesian imaging with some extensions to non-Cartesian trajectories [59].

While 3D acquisitions have gained traction in some applications, the majority of MRI acquisitions rely on 2D multislice encoding and acquisition. In the case of sequences such as FLASH/Gradient Recalled Echo, where a single slice is excited and acquired per repetition time (TR), the volume acquisition time (VAT) is directly proportional to the number of slices and the number of phase-encoding lines. Shortening the volume acquisition time is desirable not only in the interest of shorter scan time but also for applications such as body and cardiac and to reduce the temporal footprint of the acquired volume against physiological motion cycles. By exciting and acquiring N slice locations simultaneously, the acquisition time could be shortened by N folds, and is an independent encoding to phase-encoding undersampling and additionally can for sequences employing echo-trains or optimized k-space sampling be used without impacting the TE or sampling pattern. This is the underlying motivation for Simultaneous MultiSlice (SMS) Imaging. For sequences employing extended echo-trains the scan-time reduction with SMS is less straightforward and would result from either the reduction in number of slice packets (interleaves) or the duration of TR.

Let us consider the specific case of single shot (SS) MRI pulse sequences that employ a train of excitation pulses to acquire all the locations in the volume being imaged within a single repetition time (TR), and acquire the complete k-space data for the excited slice location after each RF excitation. The most widely used single-shot techniques include single-shot echo-planar imaging (EPI) and single-shot fast spin-echo (FSE) sequences. In a multishot version, a fraction of the k-space in each excited location is sampled in each TR or shot, and the data from the multiple shots are then combined during reconstruction to get the complete k-space coverage at each location. The minimum TR for such sequences would be determined by the time needed to image all the slices, in addition to a fixed block of time that might be allocated for image-contrast preparation.

The ability to excite multiple slice locations simultaneously would reduce the number of independent excitations that need to be carried out. If we consider the application of functional MRI where usually an ssEPI sequence is employed, and the contrast preparation time is minimal, simultaneously exciting and acquiring multiple slices would shorten TR (improve the temporal resolution) and allow for many more volumes or higher resolution to be acquired during a particular scan-time session. Single or multishot echo-planar diffusion-tensor imaging with many diffusion directions had previously been unfeasible in clinical settings because of the long scan times. Even though a significant block of time in diffusion imaging is allocated for contrast preparation, SMS has proved still beneficial in significantly shortening TR and scan time, improving SNR efficiency. These are a few example applications where simultaneous multislice excitation would be very beneficial.

7.2 Basics of SMS encoding

The idea of simultaneous multislice (SMS) excitation was first introduced long before the advent of parallel imaging [39,46] with the goal of improving sensitivity. With single-channel receiver coils being the norm at that time, researchers relied on phase manipulation to resolve the signal from simultaneously acquired slices. The simplest multiband (MB) excitation RF pulse can be written in the form of

$$RF_{MB}(t) = A(t) \cdot \sum_{n=1}^N e^{i\Delta\omega_n t + \varphi_n}, \quad (7.1)$$

where $A(t)$ is the complex RF waveform for a single slice, ω and φ determine the position and phase of each slice excited respectively, and N is the SMS factor. One of the initial approaches proposed Hadamard encoding of the multiband excitation pulses. In this approach the measurements are repeated N times with different phase patterns φ_n for each measurement to separate the signal from the N simultaneously excited slices. The slices are resolved on a voxel-by-voxel basis, by using the N individual images to create appropriate Hadamard-encoded combinations. This would not provide a scan-time benefit unless for certain specific multiaverage acquisition protocol. Also, signal-reconstruction schemes involving addition/subtraction from multiple measurements are generally prone to motion artifacts and additional phase errors. A variation of this approach was proposed with the Phase-Offset multiplanar (POMP) method [21], which added more flexibility over Hadamard encoding by introducing a specific k_y dependent-phase offset to the RF subpulses. By the Fourier shift Theorem, a shift in the time/spatial domain corresponds to a linear phase term in the frequency domain. Let us consider a 1D image $\mathbf{x}(y)$ with field of view FOV_y which can be shown to be the inverse discrete Fourier transform of the k -space signal $\mathbf{s}(k_y)$. A shift of y_0 in the image domain can be affected by a k_y -dependent phase offset in the frequency/ k -space domain as

$$\begin{aligned} \mathbf{x}(y - y_0) &\stackrel{\mathcal{F}}{\Leftrightarrow} \mathbf{s}(k_y) e^{-i2\pi k_y y_0}; \\ k_y = n\Delta k &= \frac{n}{N\Delta y} = \frac{n}{FOV_y}. \end{aligned} \quad (7.2)$$

In the case of POMP with two-slice excitation, a half FOV shift in the image from the second slice would be affected by a linear phase modulation of:

$$e^{i2\pi \frac{n}{FOV_y} * \frac{FOV_y}{2}} = e^{i\pi n} = (-1)^n \quad (7.3)$$

This corresponds to a sign alteration in every other measurement, which is also known as “phase chopping”. In contrast with the Hadamard approach, POMP allows for a direct reconstruction. To separate out the contributions from two simultaneously excited locations the FOV would roughly need to be extended by a factor of 2, which requires doubling the number of phase encodes during k-space data acquisition. While this method also does not provide a scan-time reduction, unless in specific protocol settings, many of the recent SMS developments draw from the POMP methodology.

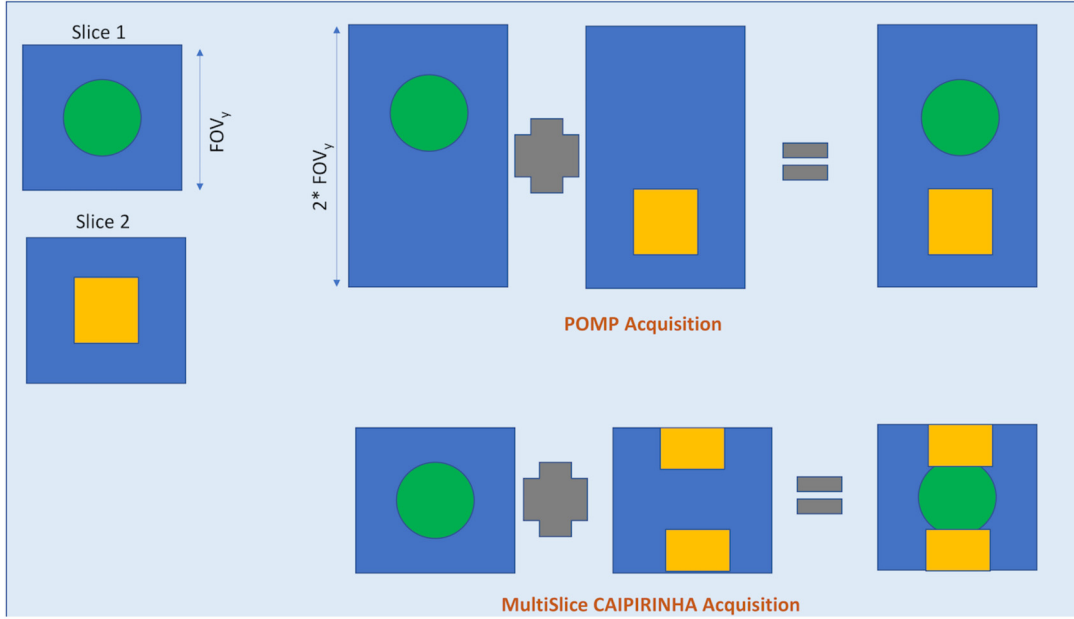
The introduction of parallel imaging [23,56,63] provided technology to overcome the inefficiencies with POMP, while also providing new insight into increasing the scan efficiency. Like POMP, multislice Controlled Aliasing in Parallel Imaging Results in Higher Acceleration (CAIPIRINHA) introduced by Breuer et al. [8] uses phase modulation, to shift the individual slices. But unlike POMP, it does not need oversampling by a factor of R because it uses the coil sensitivity information to resolve the signal from superimposed slices. In CAIPIRINHA, the FOV is still shifted between adjacent slices since this reduces the noise enhancement/improves the geometry factor of the parallel imaging reconstruction by increasing the spatial-encoding differences between locations that contribute signal to an aliased pixel. The differences in POMP and MS CAIPIRINHA acquisitions for the case of simultaneous excitation of two slices is shown in Fig. 7.1, where slice 2 has a half-FOV shift relative to slice 1.

Simultaneous MultiSlice (SMS) and MultiBand (MB) have often been used synonymously, whereas multiband terminology is also used in other applications and referring specifically to aspects of the excitations, SMS currently has a singular reference to both the excitation and reconstruction framework. For the rest of this chapter, we will refer to this technology as SMS.

When considering how to reconstruct SMS data using parallel imaging, several different techniques have been considered, similarly to the image and k-space approaches for phase-encoding undersampled parallel imaging (as detailed in Chapter 6). Both the image-space and the k-space approaches are relatively straightforward with parallel-imaging knowledge, and more easily appreciated as undersampled POMP-encoded acquisitions as illustrated in Fig. 7.1. This in particular brings back the need for using the encoding schemes from Eq. (7.3), in which for a half FOV shift sequentially phase-encoding lines are modulated by a phase modulation of the RF subpulse alternating with $[0 \text{ pi } 0 \text{ pi } 0 \text{ pi } \dots]$ between TRs.

7.3 Reconstruction of SMS using parallel imaging concepts

The SENSE framework [56] is well-suited for understanding how to reconstruct SMS data. For traditional SMS, the encoding of the individual slices in the simultaneously acquired slices is the same, and the aliasing in the simultaneously acquired slices is a direct summation of the individual slices. The GRAPPA framework [23] has also been adapted to SMS reconstruction through a progression in algorithms (SENSE-GRAPPA, RO-SENSE-GRAPPA and slice-GRAPPA) as will be discussed subsequently.

**FIGURE 7.1**

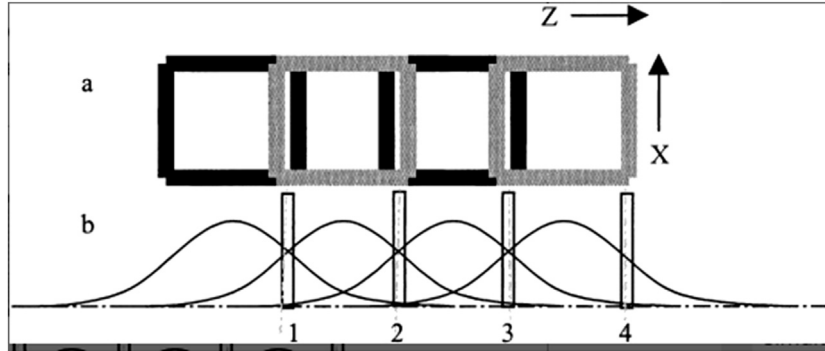
Schematic Diagram of the acquisition differences between POMP and MS-CAIPIRINHA. Two distinct slices are indicated by the circle and square respectively. Phase-offset excitation results in relative displacement of the simultaneously excited slices in the image. The FOV has been doubled in the POMP acquisition (top row) but not enlarged in the case of the MS-CAIPIRINHA acquisition, where a half-FOV shift is applied to the second slice (bottom row).

7.3.1 SENSE

The SENSE framework was introduced by Larkman et al. [32] and the schematic is shown in Fig. 7.2. If the spatial variation in sensitivity profile is known, then the simultaneous acquired signals can be considered as the solution of Eq. (7.1), where $x_1, \dots, x_m$ are the images in individual channels, $\hat{x}_1, \dots, \hat{x}_n$ the unaliased and sensitivity weighted images for the resolved slices, and C_{ij} the sensitivity profile for location i and channel j :

$$\begin{bmatrix} C_{11} & \cdots & C_{1n} \\ \vdots & \ddots & \vdots \\ C_{m1} & \cdots & C_{mn} \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \vdots \\ \hat{x}_n \end{bmatrix} = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}. \quad (7.4)$$

With known/estimated sensitivity profiles, the least squares solution for Eq. (7.4) produces the images of the individual slice locations.

**FIGURE 7.2**

First demonstration of resolving signals from SMS excitation using array coil sensitivity profiles from Larkman et al. [32]. In (a) four overlapping surface coils are shown, and in (b) magnitude sensitivity profiles for the coils at their center line and slice locations (1, 2, 3, 4) are shown. For this example the slice positions were chosen such that the magnitudes of the sensitivities of nearest neighbor coils are approximately equal.

7.3.2 Extended FOV methods

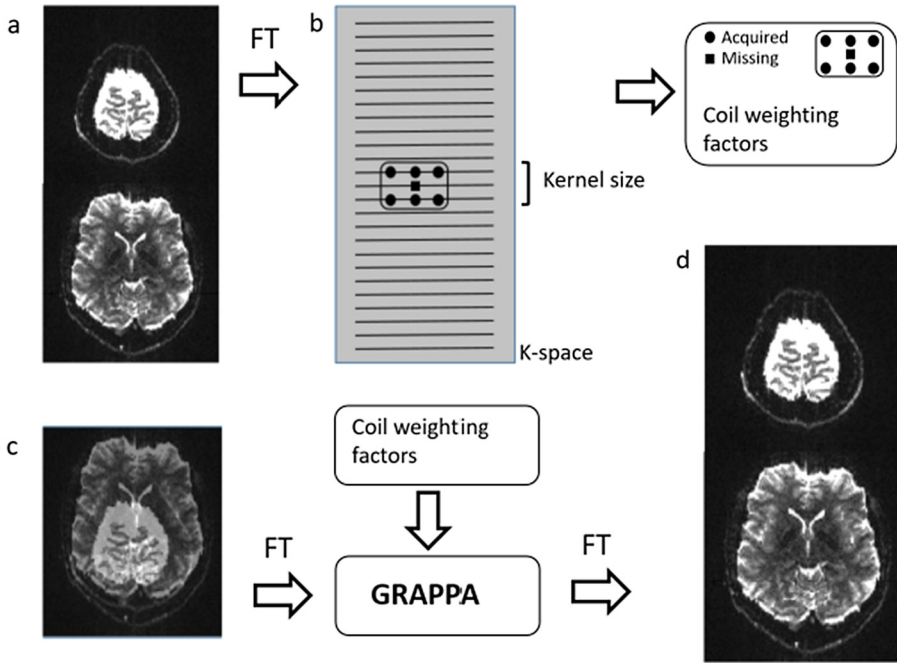
The first method, originally introduced as SENSE-GRAPPA, uses the GRAPPA framework for reconstruction, and the aliasing intuition from the SENSE framework for creating an extended FOV—the POMP acquisition framework.

7.3.2.1 SENSE-GRAPPA

The data-driven k-space approach for solving Eq. (7.4) is best understood by referring back to the POMP technique as illustrated in Fig. 7.1. In this case the acquired FOV for two slices when concatenated can be thought of as two times as large, and when only every other phase-encoding line is acquired in that concatenated FOV, the resulting image is the same as the collapsed image from two simultaneously excited slices. This approach is the concept underlying SENSE-GRAPPA. The POMP-like images used for calibration of coil-weighting factors in SENSE-GRAPPA were acquired, unlike in POMP, by separately acquiring a regular FOV image for each slice and then concatenating them to synthesize the extended FOV image. Parallel imaging allows for SMS to both have an R-fold undersampling in phase-encoding and N-fold slice-aliasing, and was tackled in SENSE-GRAPPA by solving two conventional GRAPPA problems, each with a 1D undersampling of N and R, respectively.

In the SENSE-GRAPPA method, the most common approach for reconstructing separated images is to apply the Fourier Transform (FT) along the extended/concatenated phase FOV dimension. This is then followed by partitioning the FOV into the two or more appropriate slices. A flow-diagram of the calibration data preparation and reconstruction is shown in Fig. 7.3. As an alternative approach, and again drawing on the POMP reference, after obtaining an extended k-space, the k-space signal for each slice (and then sampled at the Nyquist rate) can be obtained as

$$\mathbf{s}_{sl} = e^{-i\phi_{sl}} \left(\mathbf{s}^{ext. FOV} * e^{i\phi_{sl}} \text{Sinc}_{FOV} \right), \quad (7.5)$$

**FIGURE 7.3**

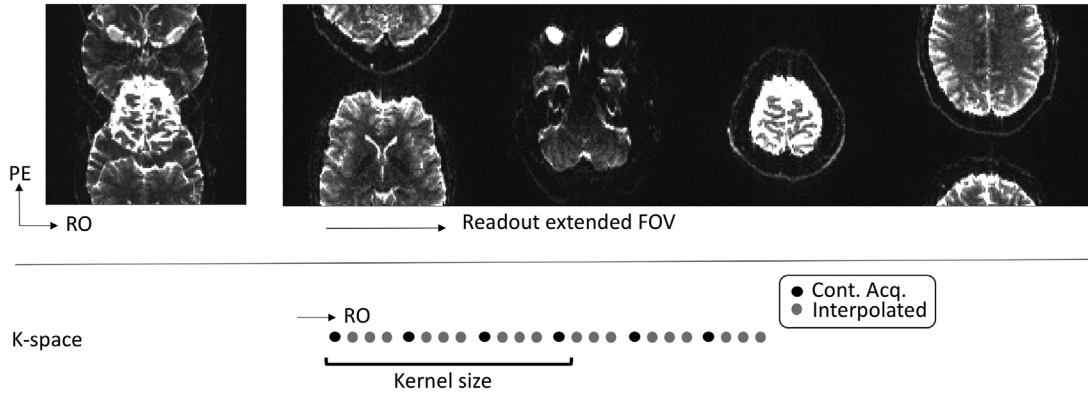
Flow diagram for the SENSE-GRAPPA calibration and reconstruction. Top row left, A, the individual slice images in calibration data have been concatenated along an extended phase FOV. In the middle, B, the standard GRAPPA kernel calibration is illustrated for the extended k-space. In the bottom row, C, SMS data for the same number of slices is shown. GRAPPA reconstruction of the SMS data in k-space yields an image, D, with the same extended FOV in image space from which the two individual slice images can be segmented.

where Sinc_{FOV} designates the sinc function defined in k-space, which is bandlimited in image space with width FOV, and ϕ_{sl} the phase for locating the bandlimited function within the extended FOV.

7.3.2.2 RO-SENSE-GRAPPA

Since the FOV in SENSE-GRAPPA is artificial, the same algorithm can be used with interchanging of the direction of the phase-encoding and the readout. This is then the RO-SENSE-GRAPPA view of the SMS reconstruction, as shown in Fig. 7.4.

This was demonstrated in the first 7T application of SMS EPI with $N \times R = 4 \times 4$ acceleration using a 16 channel coil [44]. The concatenation of the slice from SENSE/GRAPPA was altered from extending along the phase-encoding direction to be in the read-out direction, which then no longer has the same connection to POMP. This was done solely to accommodate the slice-dependent EPI ghost, and happened to be compatible with the subsequently developed blipped-CAIPI that will be described later in this chapter in connection with EPI. Whereas for in-plane parallel imaging with

**FIGURE 7.4**

Reconstruction of SMS with $N = 4$ and $1/4$ FOV shift between slices in the concatenated RO-FOV for SENSE-GRAPPA. The CAIPIRINHA shifts in the PE direction are preserved with the GRAPPA reconstruction. Separating the slices at the edges does not introduce artifacts, and the FOV shift can be corrected afterwards for each slice. In the bottom of the figure, the dimensions of the GRAPPA kernel is illustrated for the extended k-space in the readout direction.

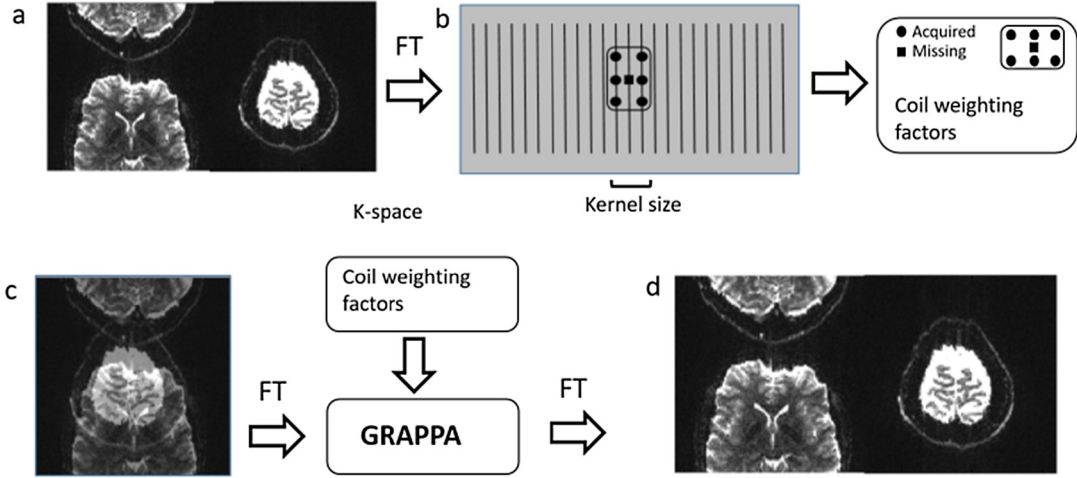
EPI, the navigator correction is applied before phase-encoding parallel-imaging reconstruction, it was applied for RO-SENSE-GRAPPA after slice-separation and before phase-encoding interpolation.

Kernel calibration

The kernel calibration for SENSE-GRAPPA and RO-SENSE-GRAPPA follow the same considerations as for GRAPPA, with the only difference being in the calibration data preparation from acquired ACS data. Following the pictorial guide of SENSE-GRAPPA from Blaimer et al. [5] as shown in Fig. 7.5, for RO-SENSE-GRAPPA, one first constructs extended FOV image from the calibration data. From this extended FOV image, a pseudo-k-space is constructed and used to compute coil weighting factors using the conventional GRAPPA technique. The obtained coil weighting factors are then applied directly to the SMS k-space, and an extended pseudo-k-space is reconstructed. Following the Fourier transform along the readout the signals can be resolved to the specific slices.

7.3.3 Slice-GRAPPA

The k-space slice separation for SENSE-GRAPPA from Eq. (7.4) forms the basis for slice-GRAPPA [61]. But in this case, instead of reconstructing an extended FOV as in SENSE-GRAPPA or RO-SENSE-GRAPPA, the data are reconstructed directly into the separated slices as depicted in Fig. 7.6a. In Fig. 7.6b, the convolution kernels for slice-GRAPPA are illustrated for reconstructing a single slice and channel-specific data-points. For each channel and slice, a unique set of convolution kernels are computed that are used to reconstruct each slice independently.

**FIGURE 7.5**

Flow diagram for the RO-SENSE-GRAPPA calibration and reconstruction. Top row, left, A, shows the calibration data from individual slices concatenated into an extended FOV along the readout direction. In the middle, B, the standard GRAPPA kernel calibration is illustrated for the artificially extended k-space. In the bottom row, C, SMS data for the same number of slices is shown. GRAPPA reconstruction of the SMS data in k-space yields an image with identical extended FOV in image space, D, from which the images of the individual slices can be separated.

Kernel calibration

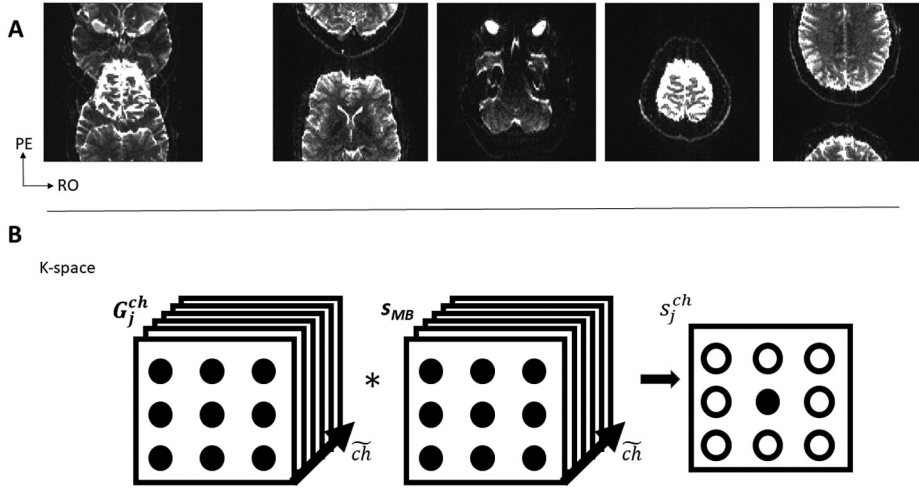
Slice-GRAPPA is a different reconstruction compared with the other GRAPPA variants. Whereas GRAPPA is used for interpolating missing data using small convolutional kernels, in slice-GRAPPA the convolutional kernels are a projection onto the solution space. For GRAPPA the reconstructed image is a combination of the measured data and the interpolated data, but for slice-GRAPPA the measured data is not used together with the reconstructed data to resolve the N simultaneously acquired slices. The coil weights G_j in slice-GRAPPA, for acquired SMS kspace s_{MB} and targeted slice resolved k-space s_j , are constructed such that

$$G_j(s_{MB}(k)) = s_j(k) \quad \forall j \forall k \quad \text{where } j = 1..N, \quad (7.6)$$

and the coil weights G_j are determined by solving for G_j in Eq. (7.6) by substituting

$$s_{MB} = \sum_j s_j. \quad (6a)$$

Pictorially, the method is illustrated in Fig. 7.7 for a kernel that independently reconstructs a single slice from the superimposed slices (the multichannel aspect is suppressed in the figure). This kernel construction was initially adapted due to its superior g-factor performance [77].

**FIGURE 7.6**

Reconstruction of SMS with $N = 4$ and $1/4$ FOV shift between slices in Slice-GRAPPA. The CAIPIRINHA shifts in the PE direction is preserved with the slice-GRAPPA reconstruction. Reconstructing each slice with a projection reconstruction does not introduce artifacts, and the FOV shift can be corrected afterwards for each slice. In the bottom line b, the construction of the convolution kernels is illustrated.

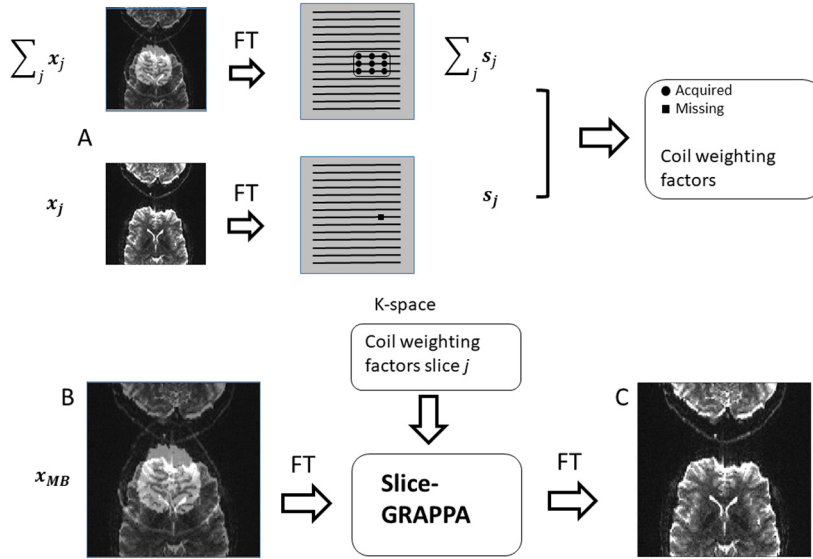
A similar SMS extension of the ARC [2,6] method of parallel imaging for reconstruction of individual slices from the superimposed slice data is known as the slice-ARC technique.

7.3.4 Split-Slice-GRAPPA

Slice-GRAPPA was introduced as part of the developments in the human connectome projects and quickly adapted in these studies because of the superior image quality. Eventually, it also revealed short-comings for residual slice aliasing and temporal SNR (tSNR), which could be improved upon with a more complex dynamic calibration techniques [42] and subsequently more elegantly with Split-Slice-GRAPPA. The latter has turned out to be similar to unbiased slice-GRAPPA in minimizing slice leakage and analogous in performance to SENSE-GRAPPA.

It was realized in [11] that the slice-GRAPPA could be trained to minimize the sensitivity to phase variations that varied significantly from the calibration phases by ensuring the MB data mapped to the proper SB data, while not mapping to alternate slices. Formally for Split Slice-GRAPPA (SpSg), the kernels were determined such that

$$G_j \left(\sum_i s_i \right) = s_j \quad \forall j \text{ AND } G_j(s_i) = 0, \quad i \neq j. \quad (7.7)$$

**FIGURE 7.7**

Flow diagram for the slice-GRAPPA calibration and reconstruction is shown. Top left, A, individual slices acquired with single-band excitation and their summation, which are used for calibration of the coil weighting factors, are shown. In the bottom row, B, SMS data for the same number of slices are shown. After slice-GRAPPA reconstruction in k-space, a resolved slice, C, is reconstructed.

Kernel calibration

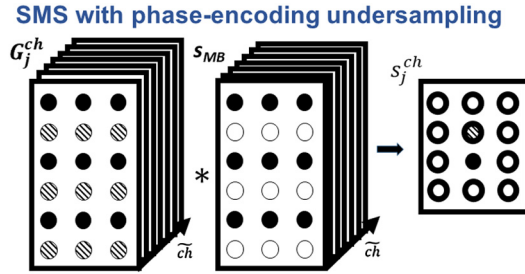
To solve for this, the system of equations was adapted to solve the penalized least squares, with $\lambda = 1$, yielding

$$\arg \min_{G_i} \left\| G_i \left(\sum_j^N s_j \right) - s_i \right\|_2^2 + \lambda \sum_{j \neq i} \| (G_i(s_j)) \|_2^2. \quad (7.8)$$

The construct with suppressing residual aliasing in slice-GRAPPA is an approach that does not have an obvious analogy for conventional parallel imaging. Nonetheless, the importance of the SpSg algorithm was quickly realized for a variety of SMS applications and adapted for large-scale projects, such as the human connectome projects [69].

7.3.5 SMS with phase-encoding undersampling

Combining phase-encoding undersampling, or parallel imaging as it is more commonly described, with SMS can provide significant scan-time reduction or other advantages such as reduction in distortion and blurring due to reduced echo-spacing in the case of single-shot acquisitions. So it is a natural extension to combine SMS with phase-encoding undersampling. In both SENSE-GRAPPA and slice-

**FIGURE 7.8**

The slice-GRAPPA convolution kernels for reconstructing SMS with phase-encoding undersampling. From the acquired k-space with R different kernels, the calculated values for R different k-space locations is simultaneously determined.

GRAPPA, this is typically accomplished by sequential application of two kernels. Firstly, the SMS signal is resolved into the channels for each slice, and then conventional GRAPPA is used for interpolating the missing phase-encoding samples. These can also be used jointly, such that the acquired signal is projected onto the entire fully resolved signal space. The slice-GRAPPA kernels calculated from Eq. (7.8) or Eq. (7.10), are constructed such that, for an R-fold phase-encoding undersampling, R slice-GRAPPA kernels are calculated from the same system of equations from a reference acquisition with Nyquist sampled phase-encoding (Fig. 7.8).

7.3.6 Reconstruction of SMS for EPI

Historically, the importance of SMS became apparent with its use in EPI functional MRI (fMRI) and Diffusion Tensor Imaging (DTI) applications. In this section, we will discuss the additional considerations for the extension of SMS acquisition and reconstruction to echo-planar imaging.

7.3.6.1 Blipped-wideband and blipped-CAIPI encoding

Since the data in single-shot EPI are collected in a single shot/TR, the RF phase-cycling approach for generating CAIPIRINHA shifts is not feasible. Instead, the slice gradient, a gradient perpendicular to the slice, is used to augment the encoding. This was first proposed by Nunes [57] using a blipped wideband approach. In the blipped wideband approach, a short blipped slice-gradient is added simultaneously with the phase-encoding blip. The gradient moment is calculated to continuously adding a slice-specific phase, such for a slice with 1/2 FOV shift, the accumulating phase is $[0, 180^\circ, 360^\circ, 540^\circ, \dots]$

This was subsequently improved by Setsompop et al. [61] using a balanced blipping approach, such that the added phase for the same 1/2 FOV shift is $[0, 180^\circ, 0^\circ, 180^\circ, \dots]$. Mathematically, there is no difference, but experimentally there is a significant difference since the added phase varies subtly across the slice. As the added phase increases, as it does in blipped wideband, the phase across the slice leads to signal loss. The distinction is illustrated in Fig. 7.9, from Setsompop et al. [61].

Instead of the slice-gradient blipping, the slice-gradient can also be modulated during each readout in the EPI echo-train [49,73,4].

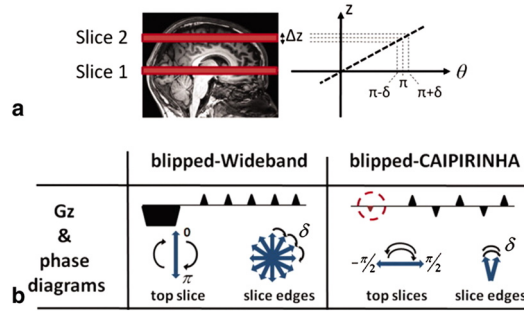
**FIGURE 7.9**

Illustration from Setsompop et al. [61] of the accumulating phase for blipped-wideband and blipped-CAIPI slice encoding.

Nyquist ghosting, one of the most common artifacts in echo-planar imaging, originates from eddy-current delays in the alternating readout gradients leading to nonalignment of the odd and even k-space lines, B0 inhomogeneity, eddy currents from slice-encoding blips, and other sources. The ghost correction becomes particularly tricky in multiband EPI since ghost correction prior to separation of the simultaneously excited slices would theoretically correct for slice-averaged ghosting only. A “tailored ghost correction” where residual ghosting after separation of slices was corrected using navigator information from the single-band calibration acquisition has been proposed [44].

Just as residual $N/2$ eddy-current ghosts are hard to separate from aliasing in parallel imaging, they complicate slice unaliasing as well. The eddy currents from opposite’s polarities of the readouts create ghosting that can normally be corrected with simple time-correction, with the linear phase estimated from the EPI navigator. While the simplest eddy-current model is a spatially and temporally invariant single timing correction, the eddy currents from gradient switching vary spatially and different slices may need different correction [3, Chapter 16.1]. The common extensions from this simplest model are slice-dependent eddy-current and 2D-eddy-current correction models. The latter is more common for oblique imaging, where two gradient elements are used simultaneously. The 2D eddy-current correction will not be considered here, but references for the more complex models are Nyquist Ghost Corrected SENSE-GRAPPA (NGC-SENSE-GRAPPA) [31] image space [37], k-space [33]. When the eddy-currents have enough spatial variation, the ghosting in the PE direction acts similarly, albeit with more subtlety, to the blipped-CAIPI shift. To avoid the interference of eddy current-induced ghosting with slice and phase unaliasing, the SENSE-GRAPPA approach can be applied with RO as the extended dimension as in RO-SENSE-GRAPPA or the slice-GRAPPA approach that reconstructs individual slices can be used. Once the slices have been resolved, for either approach, the slice-specific eddy-current correction can be applied to obtain an “artifact free” SMS-EPI reconstruction.

7.3.6.2 Slice-GRAPPA with dual polarity

In the slice-GRAPPA approach, to account for eddy currents from alternating phase-encoding, the slice-GRAPPA projection was modified to map groupings of signals that are consistent. Essentially, a unique kernel is needed to distinguish between even–odd–even phase-encoding patches and odd–even–odd phase-encoding patches, and these can be constructed as a single kernel. For example, the 3×3 kernels

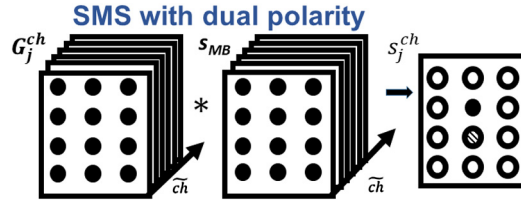


FIGURE 7.10

Illustration of the dual-polarity kernels as a single interpolation kernel.

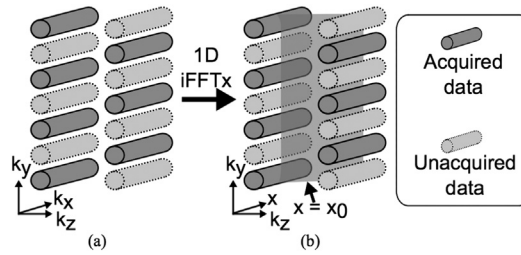


FIGURE 7.11

The hybrid-space SENSE reconstruction adopts a 3D representation of SMS acquisitions. The 3D k-space data, **a**, is transformed into the hybrid space, **b**, by a 1D inverse FFT. For each point $\mathbf{x} = x_0$, the unknown magnetization on the $\mathbf{y}$ - $\mathbf{z}$ plane is solved by $\mathbf{m}_{x_0} = \mathbf{E}_{x_0}^{-1} \mathbf{s}_{x_0}$ where $\mathbf{E}_{x_0}$ is the encoding matrix consisting of sensitivity and $\mathbf{k}_y - \mathbf{k}_z$ phase encodings, and $\mathbf{s}_{x_0}$ is a vector containing the measured signals on the $\mathbf{k}_y - \mathbf{k}_z$ plane.

were effectively extended to a 4×3 kernel as illustrated in Fig. 7.10. These extended slice-GRAPPA kernels are not shift-invariant as the k-space is normally organized, but shift-invariant for double shifts in the phase-encoding direction. The similar reasoning has subsequently been used in in-plane parallel imaging as dual-polarity GRAPPA [27,28].

7.3.6.3 SENSE-model for EPI

For parallel imaging reconstruction, there is clear link between SENSE- and GRAPPA-type reconstructions, and the same constructs holds for SMS. As discussed earlier, the SENSE model might indeed be the simplest way to consider how to separate fully sampled SMS, and it is conceptually equivalent to the Hadamard construction, where the multiple phase-modulated acquisitions have been replaced with single acquisition with multiple receivers. Both phase-encoding undersampling and the Nyquist ghost from the EPI echo train can be incorporated in the SENSE framework using a more complex encoding model. Zhu et al. [81] considered the same eddy-current model as illustrated in Fig. 7.11.

For SENSE with spatially varying eddy currents, the authors introduced the matrix-decoding for SENSE, represented as

$$s_j(x_0, k_y(n), k_z(n)) = \sum_{y=0}^{N_y-1} \sum_{z=0}^{N_s-1} x(x_0, y, z) \cdot C_j(x_0, y, z) \cdot e^{-i[k_y(n) \cdot y + k_z(n) \cdot z]} e^{i\theta(x_0, n, z)} \quad (7.9)$$

for $n = 1, 2, \dots, N_p$ and $j = 1, 2, \dots, N_c$ with N_p the number of slices and N_c the number of channels, and x_0 a fixed readout frequency. Using this formulation and relying on well-defined sensitivity profiles provided equivalent results to SpSg and was reformulated in computational efficient form by Hennel et al. [25].

7.4 Calibration and reference scans

As with in-plane parallel imaging, how to acquire calibration data, be it for deriving of coil sensitivity information in the SENSE framework or for computation of calibration weights in GRAPPA based approaches, is a key design consideration in SMS imaging. While a prescan-based coil sensitivity estimation was originally proposed for SENSE by Pruessmann et al. [56], autocalibration scan (ACS) has the advantage of sharing the same image contrast as the undersampled data [23]. The pros and cons and the applicability of various calibration schemes is discussed in detail by Griswold et al. [22], and these considerations in general hold for SMS as well.

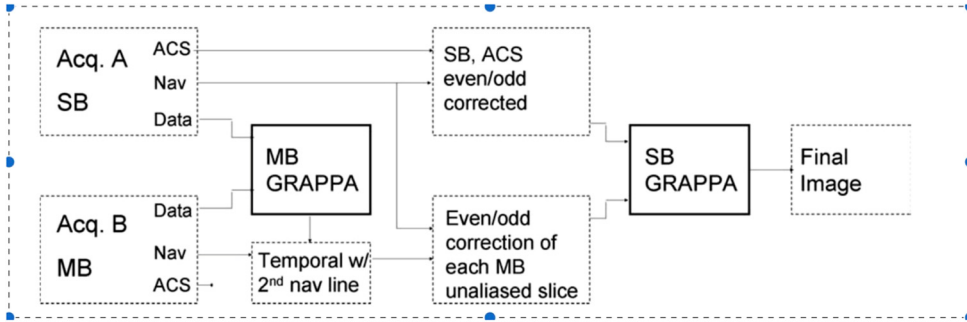
The main difference in calibration need for SMS compared to in-plane parallel imaging is that calibration information is needed from the multiple individual locations that are excited simultaneously in the accelerated acquisition. In case of SMS structural imaging, for expediency, calibration data is usually extracted from a rapid gradient-echo sequence with a conventional single-band excitation since the reconstruction methods are largely robust against differences in contrast between the imaging and calibration data. An exception to this is echo-planar sequences.

7.4.1 Calibration and reference scans for EPI

In EPI it is preferred that the calibration data has similar presence of Nyquist ghosts as the accelerated data so that these can be accounted for in the reconstruction either explicitly or implicitly. This can be achieved with a single-band EPI calibration using otherwise identical scan parameters including matched in-plane acceleration as the SMS scan, and additionally an ACS dataset that is used to compute reconstruction weights for in-plane undersampled data once the SMS data is resolved into individual slices [44] (Fig. 7.12).

Since fMRI and DTI scans include repeated volume measurements, it might be possible to spend a little more additional time on the calibration. One such approach is to acquire a fully distortion-matched and Nyquist ghosting-matched echo-planar calibration dataset by acquiring N phase cycled repetitions with the multiband excitation and R undersampled shots, where N and R are the SMS and in-plane parallel imaging-acceleration factor, respectively. In each of the phase cycled repetitions p , the phase offset between the sub-bands is varied so that they are uniformly distributed over a unit circle.

$$RF_{MB,p}(t) = A(t) \cdot \sum_{n=1}^N e^{i\Delta\omega_n(t) + \varphi_{n,p}}. \quad (7.10)$$

**FIGURE 7.12**

The flow diagram for RO-SENSE-GRAPPA for EPI with phase-encoding undersampling from Moeller et al. [44]. Each of the single-band data have an ACS acquisition and a phase-encoding undersampled acquisition in addition to a the-line navigator. The single-band phase-encoding undersampled acquisitions are used in an MB-GRAPPA (RO-SENSE-GRAPPA) reconstruction to resolved the SMS (MB) acquisition. After resolving the slice, the resolved slices are corrected for slice-specific eddy currents determined from the single-band acquisition. Lastly, a conventional GRAPPA reconstruction is used to reconstruct the full-space for each channel and each slice.

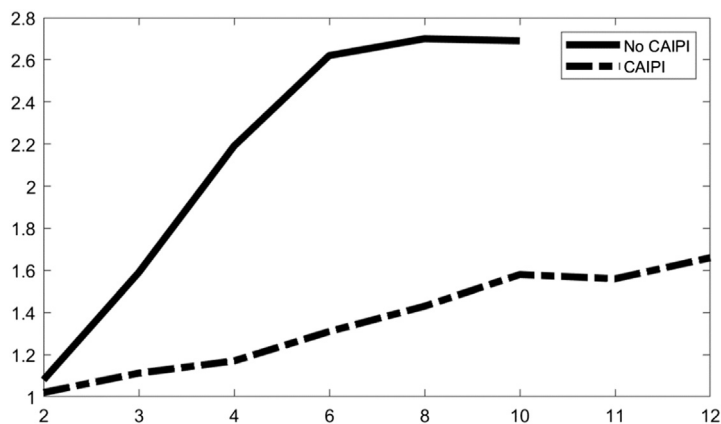
For an SMS factor of 3, the phase offset between slices would be $0, \pi/3, 2\pi/3$. The calibration data for each individual slice location can be extracted by application of the Fourier theorem to the phase-cycled multiband data, and the fully sampled data for each slice location could be collated from the individual shots. This has been demonstrated to provide a robust calibration and low ghosting levels, especially at 7T [30,64]. However, the $N \times R$ additional TRs needed for calibration acquisition would undercut the scan-time savings and would not be suited to clinical DWI and anatomical SMS imaging.

The FLEET technique (fast low-angle excitation echo-planar technique) [51,76] is a popular single-band calibration technique, which acquires all segments belonging to a slice in immediate succession rather than further apart in time, alleviating the concern of motion or breathing between segmented acquisitions. It has also been shown to exhibit low residual aliasing in SMS reconstructed images. To avoid complexities of echo-planar data altogether, fast gradient-echo sequences with low flip angles integrated in prescan are also a practical choice for calibration [66]. The optimal choice of calibration acquisition ultimately depends on the specific needs of the imaging application.

The g-factor for the k-space based methods differ based on the SNR of the ACS data. The noise in the ACS data acts a regularization term, as discussed in Ding et al. [16], and this is similar across SMS reconstruction methods. Some of these considerations are taken into account in an optimization of the hyperparameters of SMS reconstruction by Muftuler et al. [45].

7.5 Reconstruction metrics

The Signal-to-Noise (SNR) ratio is the primary measure for data quality and where SMS has an advantage over parallel imaging. With SMS, the thermal noise is shared between the N slice locations

**FIGURE 7.13**

Plot of mean g-factors from Xu et al. [77] for SMS for whole brain imaging without CAIPI and for CAIPI with 1/4 the FOV shift between slices. Without CAIPI the g-factor increases rapidly and plateaus around $N = 6$. For CAIPI the g-factor increases more gradually.

that are excited and acquired simultaneously, and there is some SNR loss associated with resolving these N slices, but there is not a $\sqrt{N}$ SNR penalty due to reduced sampling—as is the case for phase-encoding undersampling.

For completeness, we should point out that SMS will have lower SNR by a factor of $\sqrt{N}$ when compared with a fully sampled acquisition such as the Hadamard acquisition discussed earlier in the chapter, which would be composed of N individual SMS acquisitions. This $\sqrt{N}$ SNR penalty is analogous to the phase-encoding undersampling in 3D acquisitions with N -fold undersampling.

7.5.1 Noise amplification

Noise amplification in reconstruction beyond that due to reduced data measurement in k -space is quantified by the geometry factor, better known as the g-factor, in conventional parallel imaging. An analogous g-factor can be calculated for the SMS reconstruction methods previously described since it can be for any linear reconstruction. The calculation can be done with the encoding matrix for SENSE [56], Monte Carlo simulation method for linear k -space methods [58], or directly from the GRAPPA-kernels [9].

An initial evaluation for g-factors in SMS brain imaging was performed with a GRE acquisition for $\text{SMS} = 2, \dots, 12$, [77] using various CAIPIRINHA shifts. In Fig. 7.13, g-factors for adjacent slices using slice-GRAPPA without CAIPIRINHA and with a CAIPIRINHA shift of 1/4 the FOV are shown. From this figure the mean g-factor without CAIPIRINHA increases linearly up to $\text{SMS} = 6$ and then plateaus. Likewise, the g-factor with CAIPIRINHA increases linearly with increases in SMS factors, but at a slower rate. The plateau seen without CAIPIRINHA is a consequence of a few observations. For the same amount of ACS data, the linear system has more unknowns as the SMS factor increases. Furthermore as the SMS factor increases, the kernel sizes also have to increase to encode the systems better, further increasing the number of unknowns, and making the system less overdetermined. Since

the system is less overdetermined, it is easier to obtain a solution that fits well to the calibration data, but which might not be as reliable when applied to new data—typically reflected as residual aliasing. For CAIPIRINHA the same challenges of increasing number of unknowns is inherent, but the encoding model is better-conditioned, and as such it is easier to find the “right” solution. The dip for SMS = 11 and SMS = 12 leads to the question of what is being lost at the higher acceleration factors since it does not appear to be the noise amplification that is the limitation.

7.5.2 Residual aliasing

These observations are what led to considering residual aliasing as a metric, and also led to the proposition and application of SpSg. Considering the residual leakage provided the desired complementary insight. Two different approaches for this have been pursued: The simplest is to consider $\widetilde{MB}(\cdot) = [\cdots MB_j(\cdot) \cdots]$ since the combination of individual MB_j projection operators for recovering signal from slice j such that

$$\widetilde{MB} \left(\sum_j^N s_j \right) = \widetilde{MB}(s_1) + \cdots + \widetilde{MB}(s_N) = \sum_{i,j} \tilde{s}_{ij} = \sum_{\substack{i,j \\ i \neq j}} \tilde{s}_{ij} + \sum_{i=j} \tilde{s}_{ij} + \epsilon, \quad (7.11)$$

where $\tilde{s}_{ij} = MB_j(s_i)$ is the reconstruction across all channels for slice i of the input signal from slice j only. The first term is the sum of all aliased signal components, and ϵ denotes the inherent noise amplification. The original approach used a spectral analysis of the reconstructions for a slice $\hat{s}_j(t)$ determined from

$$MB_j \left(\sum_j^N \mu_n(t) s_j(\vec{k}) \right) = \hat{s}_j(\vec{k}, t) = E^H(\hat{x}_j(t)), \quad (7.12)$$

where E is the sensitivity weighted-channel combination and $\mu_n(t) = (1 + 0.1 \cdot \cos(4\pi \cdot n \cdot t))$ is chosen as a modulation function that is nonzero and resolvable with the Fourier transform. For both Eq. (7.11) and Eq. (7.12), the reconstructions have three components, the “true” resolved signal, the residual aliased signal and the noise amplification. For either method, comparing the “true” resolved signal to the residual aliased signal is elevated by the noise. One approach taken to quantitatively determine the signal leakage was to compute the “temporal” spectrum

$$\hat{x}_j(f) = F_t(\hat{x}_j(t)), \quad (7.13)$$

where μ_n defines, through construction, the frequencies f_n unique relative to the input of each slice. For a given frequency f_n , the existence/contribution of the signal in unaliased slices that are different from the input slice is then the amount of residual signal (slice leakage signal). The amount of signal leakage is quantified by the L-factor and determined from Eq. (7.13) as the largest signal in a slice distinct from the source slice. To this end, signal contributions to a slice, with a uniquely probed frequency f_n relative to other probed frequencies $f_{n'}$ are identified, masked ($\Omega = \{x_j \gg 0\}$), and summed

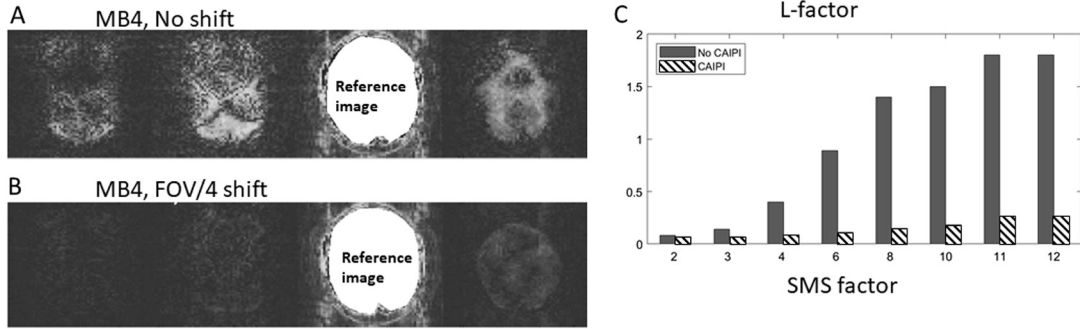
**FIGURE 7.14**

Illustration from Xu et al. [77] and adapted to grayscale for slice-leakage without, **A**, and with, **B**, 1/4 the FOV shift between slices for SMS with $N = 4$. The right figure, **C**, is a barplot of the numbers from Xu et al. [77] of the slice-leakage, and where the g-factor plateaued for $N = 6$ without CAIPI in Fig. 7.12, is when the slice-leakage effectively reached 100%.

$$L(n) = \sum_{r \in \Omega} \sum_{n' \neq n} \frac{\hat{x}_n(r, f_{n'})}{\hat{x}_n(r, f_n)} \quad (7.14)$$

and the maximum and average of $L(n)$ respectively quantified.

7.5.3 Qualitative effect of slice leakage

The qualitative effect of slice leakage, as introduced in Xu et al. [77], is shown for SMS with $\text{SMS} = 4$ in Fig. 7.14A, and quantitatively in Fig. 7.14B.

Determining how much slice-leakage is tolerable, and correspondingly which algorithm should be used for balancing noise-amplification against aliasing, can be guided by the application. Using a functional task, it was shown in Todd et al. [68] that while SpSg reduced false-positives due to residual aliasing in the SMS, it also reduced the temporal SNR as shown in Fig. 7.15. This decrease in temporal SNR correlates with an increase in g-factor for the SpSg kernels.

For rs-fMRI a detailed evaluation of SMS acquisitions with $\text{SMS} = 3$, and no CAIPI shift did not indicate any detrimental effects from residual aliasing. The level of residual aliasing as measured with the L-factor characterized these levels as 0.03 for the mean L-factor, 0.14 for the 99 percentile, and 0.44 for the maximum (e.g., in the order of 3, 14 and 44%, respectively). These levels were further increased in Xu et al. [77], to 0.05, 0.3 and 0.9, respectively, and were then used to justify the use of $\text{SMS} = 8$ with blipped CAIPI, although they indicated that higher SMS factors could be used without excessive increases in residual aliasing. In this case the Leakage factor provided a better discriminator compared with the g-factor in establishing feasible ranges for SMS. In addition to the application specific threshold in [77], the signal leakage is further reduced by using any of the unbiased methods, with an increase in the g-factor, which then can be evaluated in the classical SNR-efficiency sense.

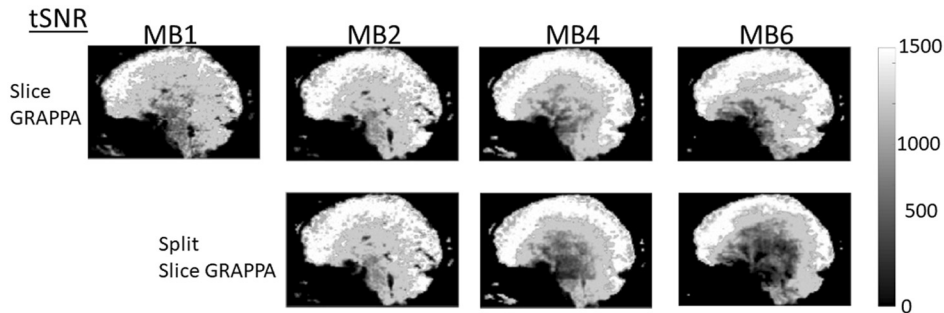


FIGURE 7.15

The effect of SMS reconstruction strategy on temporal SNR (tSNR) in fMRI adapted to gray-scale from Todd et al. [68]. The temporal SNR (tSNR) between slice-GRAPPA and split sliceGRAPPA is negligible for MB2 and small for MB4. For MB6 the difference in tSNR is larger both medially and peripherally. For fMRI the reduction in false-positives for MB6 with split slice-GRAPPA was superior to the better tSNR with slice-GRAPPA.

7.6 Extensions of SMS

7.6.1 SMS and 3D imaging

SMS can be viewed as a 3D imaging method in that it handles multislice acquisition, but it differs from most common 3D MRI sequences in the sense that it does not acquire a single contiguous volume. For 3D encoding, CAIPIRINHA uses either RF or gradient-based phase modulation to obtain (under-sampled) data with reduced effective aliasing. By considering the SMS encoding in the context of 3D encoding, the link between the first introduction of Hadamard encoding and the blipped CAIPI is more transparent, and it even makes it clearer how to extend SMS for specific applications.

Considering first Hadamard encoding for two slices, A and B, as illustrated in Fig. 7.16a, with a 180° phase modulation for the second slice. Each Hadamard acquisition is a specific k_z plane in the SMS k -space. The CAIPIRINHA sampling alternates between the two Hadamard acquisition conditions, and for each phase-encoding it only samples one of the Hadamard encoding conditions. In CAIPIRINHA both Hadamard conditions are acquired, both Hadamard conditions are undersampled, and the under-sampling pattern is shifted relative to one another. When fully sampled, the FFT can be applied through k_z , which is the same as the Hadamard combination of slice A + B and slice A–B. As a second example, in Fig. 7.16b, for $N \times R = 8 \times 2$ with a regular CAIPIRINHA shift of FOV/4, a similar seesaw sampling pattern is encoded. The pattern for which k_z phase-encodes are acquired in a SMS acquisition is reflected in how the CAIPIRINHA shifts are implemented with slice gradient variations, and, since the slices are getting closer, a larger gradient blip is necessary to impart the slice-specific phases.

The blipped-wideband and blipped-CAIPI approaches can also be considered in this construct so that the monotonic increasing phase in blipped-wideband is equivalent to acquiring on a diagonal in the SMS k -space, which is equivalent to the voxel-tilting effect [61].

Previously, a multislabs approach that divided the whole acquisition into several sub-volumes [19] and conducted a 3D encoding for each subvolume has been demonstrated for whole brain diffusion and fMRI studies [53,54]. High SNR efficiency and the achievable high resolution make these approaches appealing, though there are technical challenges from intershot phase errors varying in all three direc-

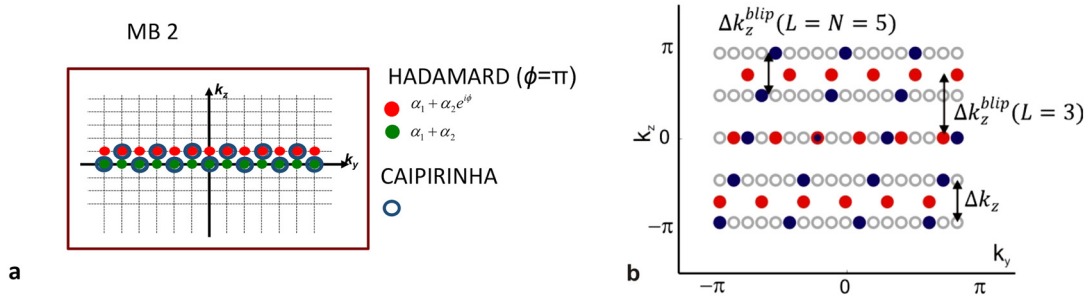


FIGURE 7.16

3D k -space and k_z phase-encoding in SMS. **a)** The phase in Hadamard encoding, and the corresponding CAIPIRINHA samples for two slices. For CAIPIRINHA, when the FFT is applied across all the phase-encoding samples, slice a_1 and a_2 are fully sampled, and the FOV of a_2 is shifted by $1/2$ the FOV when $\phi = \pi$. For **b)**, an example from Zahneisen et al. [80] of the sampling with five slices, and a FOV/3 shift between adjacent slices. Each row is the phase combination that would be obtained in one of the five acquisitions in a Hadamard acquisition. The blue (dark gray in print version) dots is the sampling for a $1/5$ FOV shift, and the red (mid gray in print version) dots are the sampling acquired when using a $1/3$ FOV shift [80].

tions and slab-boundary artifacts [17,70,75]. A combined multislab SMS approach was demonstrated, where high resolution is achieved by simultaneous reconstruction of overlapping slabs [62,72]. In the gSLIDER-SMS approach, slabs were RF encoded with a scheme similar to Hadamard to further improve orthogonality of the reconstruction bases [60]. SMSlab, a novel 3D k -space Fourier encoding and reconstruction framework proposed by Dai et al., applies joint RF and gradient encoding to a more traditional 3D multislab acquisition where interslab gaps exist [14]. Bruce et al. [10] formulated multislab multishot SMS data reconstruction in an expanded SENSE formalism that addressed both aliasing and motion correction.

7.6.2 Non-Cartesian SMS

The primary use of SMS is for Cartesian applications, but as with most encodings, it can also be used for non-Cartesian acquisitions. For most non-Cartesian parallel imaging reconstructions, the CG-SENSE framework is used [55]. For CG-SENSE any known MR specific encoding effects can be included into the reconstruction and combined with undersampled encoding when used with multichannel systems.

For SMS in non-Cartesian applications, two different classes of encoding are used: the equivalent of CAIPIRINHA and blipped-CAPI. To consider the differences in the FOV shift for non-Cartesian encoding, we briefly discuss both radial and spiral SMS.

For radial acquisitions [79], the CAIPIRINHA (or Hadamard) encoding is used for alternating spokes since each SMS spoke can be encoded with different phases. Especially for the case of two simultaneous slices, $[A, B]$, the application of alternating pairs $[0^\circ, 0^\circ]$ and $[0^\circ, 180^\circ]$ has the unique property that the signal from slice B is essentially cancelled when an NUFFT is used on the series of SMS spokes. To recover the signal from slice B, the SMS spokes are modulated by an alternating 0° and 180° phase, such that the NUFFT of the SMS signal instead cancels slice A and recovers slice B.

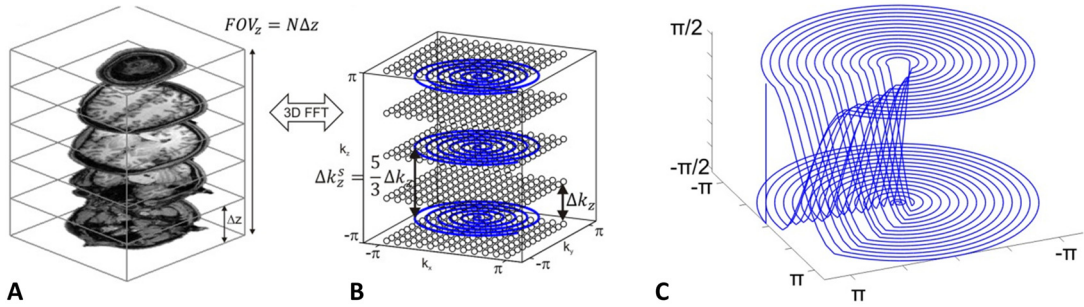
**FIGURE 7.17**

Illustration of the connection between SMS encoding in image space (a), and sampling in k-space (b) for an SMS acquisition with $N = 5$, and $1/3$ FOV between slices. Each k_z -plane can be sampled either with Cartesian or non-Cartesian sampling. (c) illustrates how to add slice-encoding blips into a single-shot spiral readout. Adapted from Zahneisen et al. [80].

This is then independent of multichannel systems, but, for more than two slices, they found that a more complete reconstruction is necessary analogous to the previous two-slice Hadamard example.

For spiral acquisitions, the blipped-CAIPI with slice-encoding blips is more amendable, although for segmented spiral acquisitions (and notably similarly for segmented Cartesian EPI), both blipped-CAIPI and CAIPIRINHA can be used. For EPI, the blipped-CAIPI slice-encoding are designed during the acquisitions dead time where the phase-encoding blips is used. For spiral acquisitions there is no inherent dead time where slice-encoding blips can be inserted. Instead, acquisition dead times have to be “created” where slice-encoding blips can be inserted. The design of the blips are chosen such that the inherent 3D-SMS k-space for Fig. 7.16b is approximately uniformly undersampled. For SMS spiral blipped-CAIPI, the k-space and sampling is illustrated in Fig. 7.17b.

For a technique such as spiral blipped-CAIPI, improvements from field monitoring have been shown to be advantageous [71]. When segmented acquisitions are used, it has been demonstrated in whole-heart myocardial perfusion imaging to be more advantageous to use a Hadamard cycling scheme [78], instead of spiral blipped-CAIPI additionally reducing the demands on the gradient performance.

Similar to the use of iterative reconstruction with constraints for calibration consistency and acquired data consistency that were previously applied to improve non-Cartesian in-plane undersampled data [36], slice-SPIRiT model has been developed to improve MB spiral cardiac MR data [65,78] and Cartesian cardiac data [15].

7.7 Applications of SMS

SMS imaging has proved to be a useful speedup tool, in addition to parallel imaging, in several clinical applications. Reports on SMS-accelerated clinical knee and spine imaging were presented by Fritz et al. [18] and Longo et al. [34], respectively. Norbeck et al. [47] applied SMS to PROPELLER [50] where they acquired the calibration data with single-band excitation from a single blade in central k-space.

From imaging contrast considerations, short TRs enabled by SMS might help improve T1 contrast. In T2 and T2 FLAIR-weighted imaging, longer TRs are needed to achieve the desired image-contrast weighting. In these applications, SMS is better utilized by extending slice coverage or acquiring thinner slices for the same slice coverage while keeping the TR same as in the original protocol.

One factor limiting the adoption of SMS technique to Rapid Acquisition with Refocusing Echoes (RARE), also known as turbo-spin echo (TSE) or fast-spin echo (FSE), which is the clinical workhorse for anatomical brain, spine and musculoskeletal imaging has been the higher peak B1 amplitude and energy of multiband pulses in a sequence that is already high in energy deposition. The use of PTx can help with reducing the increases in B1 [43]. Increasing the duration of the RF pulse would reduce peak B1 but would increase the echo spacing and have detrimental effect on image quality. Other mitigation strategies such as variable rate excitation [13] and Power Independent of Number of Slices (PINS) radiofrequency pulses [48] have associated challenges such as sensitivity to off-resonance and periodic slice profiles. Massive acceleration in brain imaging was demonstrated by Gagoski et al. [20] with SMS Wave-CAIPI and multiPINS pulses.

Comparatively, SMS has been widely adopted with EPI to enable higher direction diffusion tensor imaging in clinically feasible scan times and to enable higher temporal/spatial resolution for functional MRI.

The benefit of SMS in functional MRI is quite obvious—acquisition of more time points within a given scan time increases the statistical power. In the human connectome protocol, an SMS factor of 8 is employed for the GRE EPI fMRI protocol at 3 Tesla.

SMS can also help achieve a high enough spatial and temporal resolution in GRE EPI based Dynamic Susceptibility Contrast (DSC) perfusion techniques. An SMS multi-echo EPI sequence was used for DSC perfusion imaging with leakage correction and vascular permeability measurements [74]. SMS was demonstrated with Spin and Gradient Echo (SAGE) EPI to provide more extensive slice coverage and detailed information about tumor vasculature compared with conventional GRE DSC perfusion imaging [24].

For SE EPI Diffusion Imaging, both the excitation and the refocusing pulses need to be multiband, so typically SMS factors are limited to a factor of 3. Application of SMS has now enabled diffusion tensor imaging with a high number of directions and multiple shells in clinically feasible scan times. Furthermore, in diffusion imaging, shortening the TR below the 3000–3200 ms range at 3T for multi-slice imaging is not recommended because of T1 relaxation considerations in gray matter [43]. It needs to be noted that reduction in TR by using higher SMS factors can also lead to a reduction in the SNR of the SMS data.

Since motion poses serious challenges to SMS reconstruction, adoption of SMS in diffusion outside the brain has been limited to anatomies less affected by physiological motion, such as the breast and prostate. Segmented diffusion techniques, such as readout-segmented echo-planar imaging (rsEPI) [52] and multiplexed sensitivity-encoding (MUSE) [12], enable high-resolution diffusion, but the scan time increases proportionally to the number of shots. When SMS is extended to such techniques to reduce the TR and scan time [26,29,38], additional steps need to be considered for phase reconciliation between the segments. Multislab SMS acquisitions are generally used only in applications where submillimeter resolution is desired. McKay et al. [40] did a study in breast diffusion in which they used SMS to obtain the left-right coverage with thin-slice sagittal acquisitions and compared the axially reformatted slices against axially acquired rsEPI with five shots with comparable scan time and a faster regular ssEPI DWI acquisition. Image quality for the axially reformatted SMS acquisition was found to be

the highest. Some innovative extensions of SMS in DWI include multiband refocusing with 2D RF excitation [1] and in-plane multiband (multiplexed phase FOV) with 2D RF excitation pulse for high-resolution breast DWI [67].

SMS remains an active field, and new clinical applications employing this technology continue to emerge. In this section we have attempted to provide a few examples of its clinical utility rather than an exhaustive list.

7.8 Summary

In summary, this chapter describes the evolution of SMS from multislice Hadamard and POMP techniques that were proposed before the advent of either multicoil receiver arrays or parallel imaging. It then explains the various SMS reconstruction algorithms with special emphasis on adaptations of these methods to echo-planar imaging, lays the framework for comparison of the different reconstruction methods based on reconstruction metrics such as noise amplification and residual aliasing, and discusses a few practical design considerations and trade-offs, such as choice of calibration scan and protocol optimization in the context of imaging applications that have widely adopted SMS technology.

Similar to slice-encoding undersampling in parallel imaging (Chapter 6), SMS methods use the spatial encoding of coils to resolve aliasing between simultaneously excited slices and hence can only be supported by a receiver coil with appropriate coil geometry having separation of elements in the slice direction.

FOV shifts, commonly known as CAIPIRINHA shifts, effected by RF or gradient-phase modulation, are ubiquitously used with SMS to alter the location of aliased to improve the geometry factor and the conditioning of the reconstruction matrix. The encoding with SMS and CAIPIRINHA reduces dependency on the coil geometry and accommodates the use of higher acceleration with relatively few channels, as opposed to parallel imaging which, due to higher g-factors, generally is not used for accelerations larger than two with eight or less channel elements.

To conclude, SMS technology is a powerful speedup tool that can be used independently or in conjunction with parallel imaging to shorten scans or to accommodate more flexibility in the imaging protocol such as increased resolution or coverage.

7.9 Exercise

Workshop Tutorial from the ISMRM Workshop on Simultaneous MultiSlice Imaging: Neuroscience & Clinical Applications, Pacific Grove, CA, USA, 19–22 July 2015.

Material is created primarily by Felix Breuer, with contributions from Kawin Setsompop and Steen Moeller for the Split slice-GRAPPA and unbiased slice-GRAPPA reconstructions (<https://dx.doi.org/10.5281/zenodo.5149780>) [7].

7.9.1 Content of tutorial

Part I: SENSE

Part IIa: SENSE/GRAPPA Hybrid with ACS arranged according two cmaps in SENSE

Part IIb: SENSE/GRAPPA hybrid with extended FOVs

Part III: Slice-GRAPPA

7.9.2 Questions

The default setup is SMS with $N = 4$, and CAIPIRINHA with a $1/4$ FOV shift between adjacent slices.

- a) Construct CAIPIRINHA shifts for $1/2$, $1/3$, $1/4$ and $1/5$ FOV shifts.
 - If the FOV shifts are larger than 1, what does it mean for gradient and RF encoding respectively?
- b) For reconstruction with CAIPIRINHA encoding using either SENSE-GRAPPA or slice-GRAPPA, implement the FOV shift in k-space to correct for the CAIPIRINHA encoding.
- c) Determine how the mean and maximum g-factor vary for the SENSE-GRAPPA, slice-GRAPPA, Split slice-GRAPPA and unbiased slice-GRAPPA methods for various multiband factors and various FOV shifts.
- d) Generate and calculate the leakage factor for SENSE-GRAPPA, slice-GRAPPA, Split slice-GRAPPA and unbiased slice-GRAPPA.

Appendix 7.A Extended FOV methods for SMS

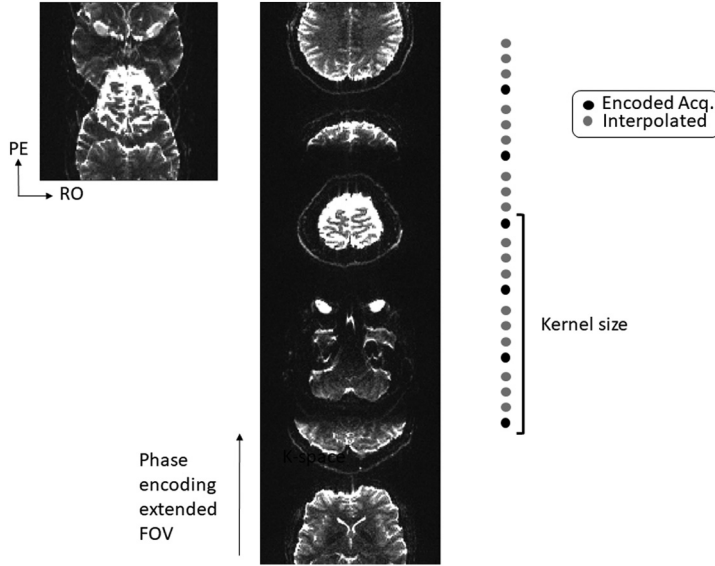
7.A.1 PE-SENSE-GRAPPA

Let us now revisit the SENSE-GRAPPA technique and pictorially consider the differences between the concepts of the different k-space techniques in the context of the CAIPIRINHA shift (the FOV shift in the PE direction for different slices). For SENSE-GRAPPA as described by Breuer et al. [8] (which we will term PE-SENSE-GRAPPA), the reconstruction aims to resolve multiple slices as shown in Fig. 7.18.

As discussed in the chapter on parallel imaging, small convolution k-space kernels can also be used as image multiplication kernels. These kernels will be smooth in the image domain since they are small in k-space they have a broader support in image-space. Likewise, since they only have a small number of Fourier coefficients with a large zero-filling component, they will also be smooth relative to the large FOV. These kernels are not well-suited to resolving the slices, when there is signal right up to the edge of the slice, such as the case when the FOV is shifted, or when it is aliased. The original introduction of CAIPIRINHA with a SENSE-GRAPPA reconstruction addressed this by adding artificial blank slices to the calibration volume, but, since this has not received much attention, that “trick” will not be further belabored in this chapter.

7.A.2 Unbiased slice-GRAPPA

The construct, with suppressed residual aliasing in SpSg, can also be achieved in a way that is more analogous to parallel imaging, using the SENSE-GRAPPA framework and a link between POMP and Hadamard encoding. The MB (or equivalently the $\sum_j s_j$ in slice-GRAPPA) is a specific subset of the extended kspace for SENSE-GRAPPA. This subset is the $1/N$ regular undersampled data from SENSE-GRAPPA. The other $(N-1)$ subsets in the extended FOV are constructions of various combinations of

**FIGURE 7.18**

Reconstruction of SMS with $N = 4$ and $1/4$ FOV shift between slices in the concatenated PE-FOV for SENSE-GRAPPA. The CAIPIRINHA shifted in the PE direction is preserved with the GRAPPA reconstruction, and separating the slices introduces artifacts at the edges of the FOV. In the bottom of the figure, the dimensions of the GRAPPA kernel is illustrated for the extended k-space in the phase-encoding direction.

the single-band in addition to $\sum_j s_j$. To extend slice-GRAPPA to include the additional information, the unbiased slice-GRAPPA is framed as solving

$$\mathbf{G}_j \left(\sum_i \alpha_i s_i \right) = s_j \quad \forall j \quad \forall \alpha_i \text{ with } |\alpha_i| = 1 \quad \text{Unbiased Slice-GRAPPA} \quad (7.A.1)$$

Practically, and since $\mathbf{G}_j$ is a linear operator, only a finite set of different combinations is necessary, and these can be created by the using α_i as the basis-vector in the Fourier matrix (or are possible, in general, in any complex Hadamard matrix). The solution is then

$$\arg \min_{\forall i} \left\| \mathbf{G}_i \left(\sum_j^N \tilde{s}_j^k \right) - \tilde{s}_j^k \right\|_2^2 \quad (7.A.2)$$

where $\tilde{s}_j^k = H(i, k) s_i$ with H the complex Fourier matrix. This is equivalent to increasing the number of equations an additional $(N-1)$ -fold.

With this formulation, Sp-Sg, Unbiased slice-GRAPPA and RO-SENSE-GRAPPA, are equivalent in performance [41] with respect to both g-factor noise and interslice leakage, and all superior to the original slice-GRAPPA in terms and accuracy and inferior in terms of noise-amplification.

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